

Aura Food: AI-Based Food Redistribution System

Comprehensive Tools, Libraries, AI Modules and Technology Stack Breakdown

SYSTEM ARCHITECTURE:	RUNTIME & APIS:	DATABASE:	SECURITY:
Decoupled SPA + Async API	Python 3.10+ / React 19	MongoDB Atlas	JWT + RBAC

1. FRONTEND & CLIENT-SIDE STACK

React 19 (v19.2.7):	Core UI library for high-speed reactive state and dashboard management.
Vite (v8.1.1):	Next-gen build tool and dev server with instant Hot Module Replacement.
Tailwind CSS (v4):	Modern utility styling engine powering responsive UI and dark mode.
React Router DOM (v7):	Declarative client routing, protected role guards, and navigation.
Axios (v1.18.1):	Promise-based HTTP client for backend REST APIs with JWT interceptors.
Lucide React:	High-performance icon system used across all navigation and modal flows.
Progressive Web App:	Service Worker & Web Manifest enabling installable mobile/desktop app.

2. BACKEND CORE & REST API ARCHITECTURE

FastAPI (v0.100+):	High-throughput async Python REST framework with auto OpenAPI docs.
Python 3.10+:	Core backend language for business rules, AI logic, and data processing.
Uvicorn (v0.22+):	Production ASGI web server handling concurrent async requests.
Pydantic v2:	Strict data schema validation, serialization, and settings management.
Python Multipart:	Processes multipart form-data for bulk inventory CSV uploads.

3. DATABASE & STORAGE LAYER

MongoDB Atlas:	Cloud NoSQL document database for Users, Inventory, Donations & Claims.
Motor & PyMongo:	Asynchronous Python driver providing non-blocking query execution.
Compound Indexing:	Optimized indexes on status, tenant_id, and timestamps for high throughput.

4. AI & SMART INTELLIGENCE MODULES

AI Expiry Predictor:	Calculates spoilage risk % using shelf life, category and storage conditions.
Food Health Score:	0-100 hygiene rating evaluating temperature, seal integrity, and freshness.
Smart NGO Matcher:	Recommends instant redistribution channels based on urgency and volume.
Impact ESG Engine:	Real-time analytics for Meals Saved, CO2 Averted, and Waste Diverted.

5. SCANNING, GEO-TRACKING & LOCALIZATION

HTML5-QRCode / ZXing:	In-browser camera Barcode & QR scanner for quick inventory check-ins.
Leaflet & OpenStreetMap:	Interactive map UI for real-time delivery route navigation & NGO tracking.
6-Language i18n Engine:	Supports English, Hindi, Kannada, Marathi, Telugu, and Urdu.

6. SECURITY, AUTHENTICATION & ROLE-BASED ACCESS (RBAC)

JWT (PyJWT):	Cryptographically signed stateless authentication tokens for API security.
Argon2 / Passlib:	Secure password hashing with salted one-way encryption.
Multi-Role RBAC:	Separate workflows for Donors, NGOs, Delivery Drivers, and Admins.

7. DEPLOYMENT & DEVOPS INFRASTRUCTURE

Vercel Platform:	Frontend PWA deployment with CDN edge caching and continuous delivery.
Render ASGI Cloud:	Containerized Python backend hosting with automatic lifecycle events.
Git & GitHub:	Source control management, branching, and automated deployment hooks.

ð fl 30-SECOND DEMO ELEVATOR PITCH FOR EVALUATORS & JUDGES

"Aura Food is an end-to-end full-stack platform built with React 19, Tailwind CSS, FastAPI, and MongoDB Atlas. It tackles the food surplus crisis by pairing intelligent AI expiry risk prediction and hygiene health scoring with in-browser barcode scanning, live delivery map routing, and 6-language regional accessibility -- ensuring that surplus food reaches hungry communities before it turns into waste."